

COMMAND NEXUS · CIT 140 · WEEK 1

Problem Brief

Meridian Forge Robotics · The Process Below

LEARNER Lacey Merced	ARTIFACT Problem Brief	STATUS Working document
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SECTION 01

Visible Failure

The incident log shows MR-17 accepted TEST_K44_START at 2:13:07 and begin repeating the test at 2:13:09. When operator sent stop request at 2:14:02, system returned ACTIVE_INSTUCTION_PENDING and repeat cycle did not stop.

SECTION 02

Verified Facts

Current scheduler: MR-17 has no active test assignment and Test K-44 does not appear in the current test catalog.

Maintenance record: Test K-44 was marked retired 186 days before the incident and removed from the active workflow.

Command telemetry: The command was validly formatted and the source label was legacy_sync.exe.

SECTION 03

Assumptions and Open Questions

Open question: Was Test K-44 still present in another service or schedule after it was retired?

Open questions: Who owns or operates the legacy_sync.exe service that sent the command?

SECTION 04

System Environment Map

SECTION 04A

People and Roles

Nia, the night QA supervisor, observed MR-17 during the incident and is helping preserve evidence.

Elias, the Robotics Systems Engineer, reviewed the command telemetry and discussed a possible controlled restart.

Mara, the production operations manager, reported that production is paused in the affected QA cell.

SECTION 04B

Processes and Rules

The current scheduler shows MR-17 has no active test assignment and Test K-44 isn't in the current test catalog. The maintenance record shows K-44 was retired and removed from the active workflow.

SECTION 04C

Information and Data

The incident log, maintenance record, current scheduler and command telemetry matter because they show what MR-17 did, the status of Test K-44 and where the command came from.

SECTION 04D

Technology and Tools

Elias Chen explains that the command was in a format MR-17 accepts. The command telemetry shows the source label was legacy_sync.exe. The telemetry does not verify who owns that service or whether it caused the incident.

SECTION 04E

Environment and Constraints

Production is paused in the affected QA cell and evidence must be preserved before attempting a restart so information isn't cleared that is needed.

SECTION 05

Investigation Boundary

Inside the Investigation

The team will examine the command source, Test K-44's remaining presence in the system and the logs leading up to this.

Outside for Now

A controlled restart is outside the investigation for now because current evidence does not show it is a verified solution and it could erase information still needed.

SECTION 06

Organizational Impact

Mara reports that production has been paused in the affected QA cell while the issue is being investigated. Mara can help with the operating schedule but the full software history still needs to be checked.

SECTION 07

Problem Statement

MR-17 is repeating a retired test command and the source of that command has not been verified. This has paused production in the affected QA cell.

SECTION 08

Next Investigation Step

Preserve and review the command logs before restarting MR-17 to determine where the command came from. Elias says a controlled restart might clear the active state but it has not been verified as a solution.

SHARED ARTIFACT

Working Document Link

<https://github.com/laceymerced/CIT140-Week-1>

WEEK 1 STOPPING RULE

Define before designing.

This Problem Brief records the problem that requires investigation. It does not recommend a solution.